

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 09

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

A $-\frac{3}{4}$

B $\frac{4}{3}$

C $-\frac{4}{3}$

D $-\frac{3}{5}$

E $\frac{3}{4}$

F $-\frac{4}{5}$

2 A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.

A $(5, 5)$

B $(2, 3)$

C $(4, 3)$

D $(6, 8)$

E $(3, 4)$

F $(3, 3)$

3 Solve $\log_2 (x^2 - 4) - \log_2 (x - 2) = 3$.

A $x = 2$

B $x = 6$

C $x = -10$ or $x = 6$

D $x = 8$

E $x = 10$

F $x = 4$

4 A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

I Complete the square: $x^2 + 2x + 3 = (x + 1)^2 + 2$.

II $(x + 1)^2 \geq 0$ for all real x .

III So $(x + 1)^2 + 2 \geq 2 > 0$ for all real x .

IV Hence the equation has no real solutions.

Which of the following is correct?

A The proof is completely correct.

B The first error is on line I.

C The first error is on line II: $(x + 1)^2$ can be negative.

D The first error is on line III.

E The first error is on line IV: a positive expression can still equal zero.

- 5 x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

6 Seven consecutive integers are written down. Which of the following statements is/are true?

- I** Their sum is divisible by 7.
- II** Their product is divisible by 7.
- III** Their sum is odd.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

7 A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

A The first error is on line I: dividing by x discards the solution $x = 0$.

B The solution is completely correct.

C The first error is on line II.

D The first error is on line I: $x^2 \div x = 2x$.

E The first error is on line I: it should give $x = 2$.

8 Which of the following statements is/are true?

I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.

II A 10% increase followed by a 10% increase is a 21% increase overall.

III If A is 25% more than B , then B is 20% less than A .

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 9 Which of the following is the negation of the statement '**every student in the class scored more than 50**'?
- A At least one student in the class scored more than 50.
 - B Every student in the class scored 50 or less.
 - C Every student in the class scored less than 50.
 - D At least one student in the class scored 50 or less.
 - E No student in the class scored more than 50.

- 10** Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?
- A** $n = 3k + 1, n = 3k + 2$
 - B** n even, n odd
 - C** n positive, n zero, n negative
 - D** $n = 3k, n = 3k + 1, n = 3k + 2$
 - E** n prime, n not prime

- 11** For real numbers a, b : $P: 'ab = 0'$; $Q: 'a = 0'$. Which one of the following is true?
- A** P is necessary and sufficient for Q.
 - B** P is necessary but not sufficient for Q.
 - C** P is sufficient but not necessary for Q.
 - D** P is neither necessary nor sufficient for Q.

- 12** *P*: 'a solid cylinder and a solid cube have equal volumes'. *Q*: 'the cylinder and the cube have equal surface areas'. Which one of the following is true?
- A** *P* is necessary and sufficient for *Q*.
 - B** *P* is necessary but not sufficient for *Q*.
 - C** *P* is sufficient but not necessary for *Q*.
 - D** *P* is neither necessary nor sufficient for *Q*.

- 13 Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?
- A Any three integers include one that is a multiple of 3, so the sum is a multiple of 3.
 - B Let the integers be $n - 1, n, n + 1$. Their sum is $3n$, which is divisible by 3.
 - C Let the integers be $n, 2n, 3n$. Their sum is $6n$, which is divisible by 3.
 - D The middle integer is the mean of the three, so the sum is three times an integer... but this needs the middle integer to be a multiple of 3.
 - E $1 + 2 + 3 = 6$, $2 + 3 + 4 = 9$ and $3 + 4 + 5 = 12$ are all divisible by 3.

14 x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

A 0

B 6

C 2

D 1

E 3

F 4

- 15** Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?
- A** The statement is true, and so is its converse.
 - B** The statement is true, but its converse is false.
 - C** The statement is false, but its converse is true.
 - D** The statement and its converse are both false.

- 16** The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

I $a \leq 3$

II $c \geq 5$

III $b = 4$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

17 $p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I** $p(x) = 0$ has at least one real root.
- II** $p(x) = 0$ has at most four real roots.
- III** The graph of $y = p(x)$ has at least one turning point.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 18** Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

19 Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

I $p(x)$ has three distinct real roots.

II $(x - 4)$ is a factor of $p(x) - 6$.

III $p(x) > 0$ for all $x > 3$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

20 Consider the statement S : **for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$** . Which of the following statements is/are true?

I S is true.

II The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.

III The contrapositive of S is true.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

END OF TEST